

Apache Log Investigation

Objective:

The goal of this assignment was to explore and analyze a simulated Apache web server log file using Splunk. Through this hands-on task, the objective was to gain practical experience in ingesting log data, extracting useful fields, performing searches, conducting statistical and time-based analysis, and visualizing results through dashboards. This exercise provided insights into real-world web server behavior and potential security issues.

Dataset Description:

The dataset used was a simulated Apache access log file containing 1000 entries. Each log entry included key information such as:

- Client IP address
- Timestamp of the request
- HTTP method (GET/POST)
- Requested resource (URL)
- Response status code (e.g., 200, 404)
- Size of the response in bytes

Key Tasks Performed:

◆ Data Ingestion:

The Apache log file was successfully uploaded to Splunk and indexed under a custom index named `apache_logs_1`. Splunk's `apache:access` source type was used to auto-parse the log structure.

◆ Field Extraction:

Manual field extractions were performed to ensure all necessary data fields were available. The following fields were extracted and named:

- `clientip`: IP address of the client

- **timestamp**: Date and time of the request
- **method**: HTTP method (GET/POST)
- **uri_path**: Requested resource
- **status**: HTTP response code
- **bytes**: Response size

♦ **Search and Analysis:**

Various SPL (Search Processing Language) queries were executed to analyze the data:

- Total number of requests
- Top requested URLs
- Most common HTTP response codes
- Number of unique client IPs
- Average size of responses
- Request volume over time

♦ **Security Insights (Advanced Analysis):**

- Identified IPs with an unusually high number of requests
- Detected frequent 404 (Not Found) errors that may indicate scanning or probing
- Analyzed GET vs POST request usage

♦ **Visualization and Dashboard Creation:**

A custom Splunk dashboard was created with multiple panels, including:

- Timechart of total request volume

- Pie chart of response code distribution
 - Table of top client IPs
 - Bar chart of most requested resources
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